

IC: <Harsith902>

My Enigma no.

$$E = \bullet 3706$$

IC: c_cforchocolate.

$$\text{My } N = 30 + \text{Remainder} \left(\frac{3706}{70} \right)$$

$$= 30 + 66 = \underline{\underline{96.}}$$

1. Guess: $(96, 5)$. would take most steps.

2.

(a) ~~$(10, 3) \rightarrow (3, 1)$~~
 $(10, 3)$

$$10 = 3 \times 3 + 1$$

$$3 = 1 \times 3 + 0$$

$$2 = 1 \times 2 + 0$$

$$1 = 2 + 0$$

$$\underline{\underline{\text{gcd} = 1.}}$$

(b) $(8, 5)$

$$8 = 1 \times 3 + 5$$

$$5 = 1 \times 3 + 2$$

$$3 = 1 \times 2 + 1$$

$$2 = 2 \times 1 + 0$$

$$1 = 1 \times 1 + 0$$

$$\text{gcd} = 1$$

steps = 6.

(2) (13, 8)

$$13 = 8q + r \quad (1) \quad (5)$$

$$8 = 5q + r \quad (1) \quad (3)$$

$$5 = 3q + r \quad (1) \quad (2)$$

$$3 = 2q + r \quad (1) \quad (1)$$

$$2 = 1q + r \quad (1) \quad (1)$$

$$1 = 1q + r \quad (1) \quad (0)$$

gcd = 1

(1) (17, 7)

gcd = 1

$$17 = 7q + r \quad (2) \quad (5)$$

$$7 = 5q + r \quad (1) \quad (2)$$

$$5 = 2q + r \quad (2) \quad (1)$$

$$2 = 1q + r \quad (1) \quad (1)$$

$$1 = 1q + r \quad (1) \quad (0)$$

steps = 5

(e) (21, 13)

$$21 = 13q + r \quad (1) \quad (8)$$

$$8 = 5q + r$$

same process.

* Table.

	pair	gcd	steps.
(a)	(10, 3)	1	4
(b)	(8, 5)	1	5
(c)	(13, 8)	1	6
(d)	(17, 7)	1	5
(e)	(21, 13)	1	7

~~(34, 2)~~ (1,1) (2,1) (3,2) (5,3) (8,5) (13,8) (21,13) (34,21)
89 (55,34) (89,55)

Que 3. The pair (89, 55) is the slowest step as it takes.

$$89 = 55 + 34$$

$$55 = 34 + 21$$

$$34 = 21 + 13$$

$$13 = 8 + 5$$

$$8 = 5 + 3$$

$$5 = 3 + 2$$

$$3 = 2 + 1$$

$$2 = 1 + 1$$

~~$$1 = 0 + 1$$~~

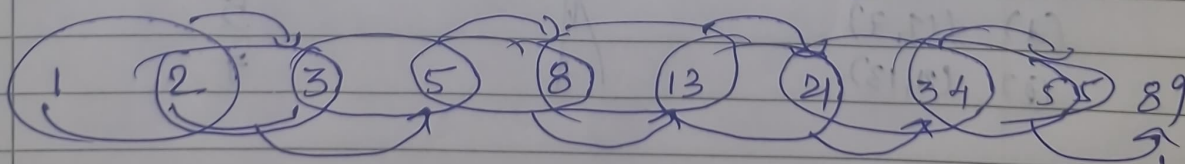
$$1 = 0 + 1 \leftarrow \text{last one.}$$

last

9 steps.

Que 4. Explanation

If we look at the examples that we did or the one which had most steps it follows a specific pattern. (The quotients).



As per the observation when consecutive pair of Fibonacci series is ~~not~~ taken it has the highest steps.

Ans Ends here. This is pattern but my logic / My observations

⇒ In Fibonacci series the quotient is always 1

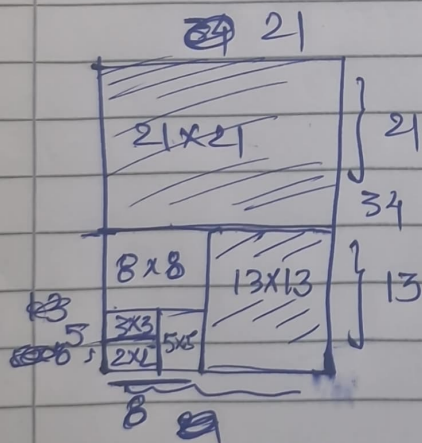
My explanation
(~~that is~~ like behind)
Fibonacci series.

Eg: $13 = 8 \times 1 + 5$, $21 = 13 \times 1 + 8$
(=1) (1)

$34 = 21 \times 1 + 13$
(=1)

so basically it ~~seems~~ increases the steps of dividing the nos to find the gcd.

like



so when we divide the squares.

then it manages to

acquire only one square among bigger Rectangle.

(my explanation is a bit off).

But like it doesn't acquire 2 big squares when we ~~only~~ to find gcd.